

Total No. of Questions : 6]

SEAT No. :

[Total No. of Pages :2

P4954

BE/In Sem. - 18
B.E. (Mechanical)
TRIBOLOGY

(Elective - I) (Semester - I) (402044 B) (2012 Course)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Write Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data, if necessary.*

- Q1)** a) Define Lubricant. How lubricants are classified? Write at least four functions of lubricant. **[6]**
- b) State the importance of recycling of used oil. How stepwise it takes place? **[4]**

OR

- Q2)** a) Explain the following properties of lubricant. **[6]**
- i) Viscosity
 - ii) Effect of pressure on viscosity
 - iii) Effect of temperature on viscosity
 - iv) Oiliness
 - v) Pour point
 - vi) Demulsibility
- b) What do you mean by additives? Write the use of EP additives and Emulsifier. **[4]**

- Q3)** Derive the expression for coefficient of friction due to adhesion " f_a " according to Modified Adhesion [Junction Growth] Theory. **[10]**

OR

P.T.O.

Q4) a) Define wear. Give the classification of wear including major and minor wear. [5]

b) Explain at least five factors affecting wear rate. [5]

Q5) Derive Petroff's equation for hydrodynamic journal bearing. State the conditions under which Petroff's equation can be used. Write its two limitations. [10]

OR

Q6) a) Write only two dimensional Reynolds equation and state the meaning of each term in equation. Write any four assumptions made for Reynolds equation. [6]

b) Explain with sketch, the mechanism of pressure development in hydrodynamic thrust bearing. [4]

